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Theme

**Development of the “QONNEXA” Web Platform for
Managing Queues in AGB (Gulf Bank Algeria).**

Presented and defended by:

CHERFI CHAHINEZ

CHIBAH ZAHRA ISRA

Defended on /26 , the jury composed of:

Dr Examiner

Dr Supervisor , Doctor at the University of Algiers 1

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Summary

Exploring the challenges faced in banking operations, we designed and developed a web-based queue management system for the Algerian Gulf Bank (AGB). Many branches still face long waiting times, lack real-time monitoring, and suffer from inefficient customer flow, which affects both client satisfaction and staff productivity. To address these issues, our system provides ticket creation, real-time queue tracking, and dashboards for employees and administrators. The development process followed standard web application steps, including requirements analysis, UML system modeling, and implementation using modern web technologies. By testing the platform in a real banking environment, we validated its effectiveness and demonstrated improvements in service organization and efficiency. Keywords: queue management, web application, banking, real-time system, customer service

Résumé

Explorant les défis rencontrés dans les opérations bancaires, nous avons conçu et développé un système web de gestion des files d'attente pour la Banque du Golfe Algérienne (AGB). De nombreuses agences connaissent encore de longs temps d'attente, un suivi en temps réel insuffisant et une gestion inefficace du flux des clients, ce qui affecte à la fois la satisfaction des clients et la productivité du personnel. Pour remédier à ces problèmes, notre système permet la création de tickets, le suivi en temps réel des files et des tableaux de bord pour les employés et les administrateurs. Le processus de développement a suivi les étapes classiques d'une application web, incluant l'analyse des besoins, la modélisation du système avec UML et l'implémentation avec des technologies web modernes. En testant la plateforme dans un environnement bancaire réel, nous avons validé son efficacité et démontré des améliorations dans l'organisation du service et l'efficacité opérationnelle. Mots clés : gestion des files d'attente, application web, banque, système temps réel, service client.

INTRODUCTION

Dans le secteur bancaire algérien, la gestion des flux clients et des files d'attente représente un défi majeur pour la qualité de service. Les agences bancaires, notamment l'AGB (Gulf Bank Algeria), font face quotidiennement à une affluence importante, générant des temps d'attente prolongés et une expérience client souvent dégradée. Cette situation engendre non seulement l'insatisfaction des clients, mais aussi une charge de travail accrue pour les employés, et une image de marque impactée pour l'établissement.

Face à ce constat, la transformation digitale des services bancaires s'impose comme une nécessité. Les clients d'aujourd'hui, y compris dans le contexte algérien, sont de plus en plus connectés et exigent des solutions modernes leur permettant de gagner du temps et de maîtriser leur parcours au sein de l'agence. C'est dans ce contexte que s'inscrit notre projet de fin d'études, visant à proposer une solution innovante pour optimiser la gestion des files d'attente à l'AGB.

L'objectif général de ce travail est donc de concevoir et développer une plateforme web dénommée "QONNEXA" dédiée à la gestion intelligente des files d'attente à l'AGB.

Le présent mémoire s'articulera autour de deux grandes parties. La première partie, théorique, présentera les fondements conceptuels de notre travail, incluant les notions de gestion de flux, les systèmes de files d'attente existants et les technologies web mobilisées. La seconde partie, pratique, détaillera l'ensemble du processus de réalisation de la plateforme QONNEXA, depuis la conception jusqu'à la validation, en passant par le développement et les tests.

INTRODUCTION(eng)

In the Algerian banking sector, managing customer flow and queues represents a major challenge for service quality. Bank branches, particularly AGB (Gulf Bank Algeria), face significant daily crowds, leading to long waiting times and often a degraded customer experience. This situation not only generates customer

dissatisfaction but also an increased workload for employees and a negative impact on the institution's brand image.

Given this observation, the digital transformation of banking services has become a necessity. Today's customers, including in the Algerian context, are increasingly connected and demand modern solutions that allow them to save time and manage their journey within the branch. It is in this context that our final-year project takes place, aiming to propose an innovative solution to optimize queue management at AGB.

The overall objective of this work is therefore to design and develop a web platform named "QONNEXA" dedicated to intelligent queue management at AGB.

This thesis will be organized around two main parts. The first, theoretical part, will present the conceptual foundations of our work, including the notions of flow management, existing queue systems, and the web technologies used. The second, practical part, will detail the entire process of creating the QONNEXA platform, from design to validation, including development and testing.

PROBLÉMATIQUE

La problématique centrale à laquelle ce mémoire cherche à répondre est la suivante : Comment améliorer l'expérience client et optimiser la gestion des flux au sein des agences AGB grâce à une plateforme web de gestion de files d'attente ? Cette question met en évidence une double difficulté : d'une part, la nécessité de réduire l'attente physique des clients en leur offrant la possibilité de suivre la file même à l'extérieur de l'agence, et d'autre part, l'importance de digitaliser le processus de prise en charge pour les rendre plus autonomes et mieux informés.

Pour atteindre cet objectif général, plusieurs objectifs spécifiques sont poursuivis :

- Automatiser la gestion des files d'attente et orienter les clients selon le type de service demandé ;
- Proposer aux clients un système de tickets à la fois physique et virtuel accessible via QR code ;

- Permettre aux clients de suivre l'évolution de la file d'attente à distance, même lorsqu'ils sont à l'extérieur de l'agence, et les notifier lorsque leur tour approche ;
- Offrir aux clients VIP la possibilité de prendre rendez-vous depuis leur domicile via la plateforme ;
- Fournir aux administrateurs un tableau de bord complet pour visualiser les indicateurs clés (temps d'attente, clients servis, satisfaction, statistiques) et gérer les tours (annulation, réorganisation) ;
- Surveiller les files et les guichets via un module de détection de problèmes basé sur des règles simples et les retours de satisfaction clients, avec affichage des alertes en temps réel.

Pour mener à bien cette recherche, nous adopterons une méthodologie en deux phases. Une première phase théorique consistera à explorer les concepts fondamentaux liés à la gestion des files d'attente, aux systèmes de tickets et aux technologies web. Une seconde phase pratique sera consacrée à la conception, au développement et aux tests de la plateforme QONNEXA, en détaillant l'architecture, les choix technologiques et les fonctionnalités implémentées.

Partie Theorique

Introduction

CHAPITRE 1 : QUEUE MANAGEMENT SYSTEMS AND CUSTOMER SERVICE IN BANKING

1.1 Introduction to Queueing Systems

A queueing system is a mathematical model that describes the process where customers arrive, wait in line, receive service, and then leave. According to recent research, every queueing system consists of three main components: the arrival process, the queue itself, and the service process [1]. The arrival process defines how customers enter the system, the queue represents the waiting line when servers are busy, and the service process determines how customers are handled and released [1]. Understanding these fundamental elements is essential for analyzing and improving waiting lines in any organization, including banks.

1.2 Basic Components of a Queueing System

Several key components determine the behavior of a queueing system. The arrival rate (λ) represents the frequency at which customers join the queue, often following a Poisson distribution [1]. The service rate (μ) indicates how quickly servers can process customers, typically following an exponential distribution [1], [2]. The queue discipline defines the order in which customers are served, with FIFO (First In, First Out) being the most common in banking [2]. The number of servers (single or multiple) also significantly impacts waiting times. When the arrival rate exceeds the service rate, queues grow longer and waiting times increase [1], [2].

1.3 Queue Configurations

Queueing systems can be organized in different configurations depending on the number of servers and queue lines. The single-server queue (M/M/1) involves one line and one server, commonly found in small shops [1]. The multi-server queue (M/M/k) uses a single line feeding multiple servers, which is the typical configuration in bank branches [1]. Some systems use multiple queues with multiple servers, where each server has its own line. Banks generally prefer the single-line, multi-server configuration because it ensures fairness through the FIFO principle and reduces customer anxiety about choosing the "wrong" line [1].

1.4 Performance Indicators of Queue Systems

To evaluate the efficiency of a queueing system, several performance indicators are used. Average waiting time measures how long customers wait before being served [3], [4]. Queue length indicates the number of customers waiting at any given time [4]. Server utilization shows the percentage of time servers are busy processing customers [5]. The abandonment rate represents customers who leave the queue before being served [4]. These metrics help bank managers identify problems and make informed decisions about resource allocation, such as opening more counters during peak hours [3], [4], [5].

1.5 Application of Queueing Systems in Banking

In the banking sector, queue management directly affects customer satisfaction. A study by Firmansyah and Saputra (2021) demonstrated that the implementation of a queue system has a positive and significant effect on customer satisfaction [6]. In the Algerian context, research at BNA Ouargla showed that inadequate ATM availability leads to longer waiting times, and adding one additional ATM could significantly improve service quality [7]. Traditional banks face challenges such as long waiting times during peak periods, lack of real-time queue information, and customer frustration [6], [7]. Digital solutions like QONNEXA address these issues by offering virtual ticketing, remote queue tracking, and real-time notifications, transforming the waiting experience and improving overall service quality. The theoretical performance indicators discussed here highlight the limitations of traditional queues. The following chapter will explore how modern digital queue management systems address these challenges.

RÉFÉRENCES DU CHAPITRE 1

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Source

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Chapter 2 — Digital Queue Management and Smart Banking SystemsBanking

2.1 Traditional vs Digital Queue Management

2.1.1 Traditional Queue Management Systems

Traditional queue management in banks is mainly based on manual methods such as physical waiting lines or numbered ticket systems. In the waiting line model, customers stand in line and are served in arrival order. In the ticket system, customers receive a number and wait until it is called [1], [2].

[Insert Figure 2.1: Traditional paper ticket dispenser in bank branch]

Although these methods organize customer flow, they provide no information about waiting time and offer limited control over service organization [3].

2.1.2 Digital Queue Management Systems

Digital queue management systems use technological tools to manage customer flow more efficiently. Customers can obtain tickets through kiosks, web platforms, or mobile applications, while the system automatically manages queue order and provides real-time information about waiting time [1], [5].

These systems also collect operational data that helps organizations improve service performance and resource allocation [4], [6].

2.1.3 Comparison Between Traditional and Digital Systems

Table 2.1 – Traditional vs Digital Queue Management

<u>Aspect</u>	<u>Traditional</u>	<u>Digital</u>
Queue organization	Manual	Automated
Waiting time info	Not available	Real-time
Communication	Limited	Displays / notifications
Data collection	None	Operational analytics

Source: Adapted from [3], [4]

2.2: Smart Queue Management Systems

2.2.1 Electronic Ticketing

Electronic ticketing systems allow customers to receive digital queue tickets via SMS on their mobile phones. When a predetermined number of tickets remain until the user's turn, an SMS reminder is automatically sent [7]. This enables customers to leave the bank and run errands while waiting. The system also supports printing paper tickets for those without mobile phones or who have difficulty using them [7].

Tickets can be customized according to service type and priority level. Check-in methods include self-service kiosks with touchscreens, mobile applications, and web portals [8].

[Insert Figure 2.2: Customer using self-service ticketing kiosk]

2.2.2 Queue Displays

Queue displays use LED screens or digital signage to broadcast real-time queue updates. LED screens show large, bold current ticket numbers (e.g., "Now Serving A024"), counter assignments ("Counter 3"), and estimated wait times ("ETA: 12 min") [9]. Digital signage can additionally display video ads or instructions.

Content management via cloud software allows remote updates, enabling managers to schedule announcements, switch languages, and customize layouts for multiple queues [9].

[Insert Figure 2.3: Digital display showing current queue status]

2.2.3 Automatic Call Systems

Automatic call systems notify customers through SMS alerts, mobile app push notifications, or display-based announcements when their turn arrives [10]. Call sequencing follows First-In-First-Out (FIFO) logic with priority overrides for VIP customers.

These systems integrate with staff terminals where operators confirm calls with a button press, triggering the next notification while updating central dashboards in real-time [10].

[Insert Figure 2.4: Staff dashboard for queue management]

2.2.4 Analytics and Reporting

Analytics modules track key metrics including average wait times, service durations per counter, and abandonment rates (customers leaving before service) [10]. Reporting dashboards present this data through real-time graphs and historical trends.

Performance insights reveal peak-hour bottlenecks and optimization opportunities like adding counters during surges, enabling data-driven decisions to improve service quality [10].

[Insert Figure 2.5: Analytics dashboard showing queue metrics]

2.2.5 Impact on Waiting Time and Organization

The integration of these technologies produces measurable improvements. Electronic ticketing eliminates confusion about service order. Digital displays provide transparency that reduces customer anxiety. Automatic notifications allow customers to wait elsewhere, reducing physical congestion [7]. Analytics enable continuous optimization by revealing peak periods and service bottlenecks [10].

2.3 Web Platforms in Banking Services

2.3.1 Digital Transformation in Banking

The banking sector has undergone significant digital transformation driven by customer expectations for convenience and remote access. Web platforms now serve as essential channels for service delivery, complementing traditional branch networks [3]. Customers expect to conduct banking business through channels that fit their lifestyles, whether visiting a branch or using online services [4].

2.3.2 Integration with Queue Management

Web platforms enable customers to interact with queue management systems remotely before arriving at the bank. Through secure websites or mobile portals, users can check current queue status, obtain digital tickets, and receive estimated wait times [5]. This pre-visit functionality allows customers to plan their arrival and reduces uncertainty about waiting times [3].

2.3.3 Appointment Scheduling

Many web-integrated queue systems offer appointment scheduling for complex transactions requiring specialized assistance. Customers can book specific time slots through the bank's website, and the system integrates these appointments with walk-in traffic management [4]. Scheduled customers receive priority while maintaining fair treatment for others, improving service organization and reducing branch congestion [5].

2.4 QR Code Technology and Mobile Interaction

2.4.1 QR Code Fundamentals

QR (Quick Response) codes are two-dimensional barcodes that store information readable by smartphone cameras. Modern mobile devices include built-in scanning capability, allowing users to access digital content instantly without manually entering website addresses [4]. This technology creates a seamless bridge between physical environments and digital services [5].

2.4.2 Applications in Queue Management

In queue management systems, QR codes serve multiple functions. Placed at bank entrances, they enable contactless check-in where customers scan to join the queue without touching shared surfaces [3]. This became particularly valuable during the COVID-19 pandemic for maintaining hygiene and social distancing [6].

QR codes can also link to service information pages, allowing customers to review required documents or procedures while waiting. Post-service, QR codes on receipts connect to satisfaction surveys, making feedback collection immediate and convenient [3].

2.4.3 Mobile Interaction Benefits

The combination of QR codes and mobile devices creates a flexible queuing experience. Customers receive real-time updates about their queue position regardless of location, allowing them to wait elsewhere productively [6]. This reduces physical congestion in banking halls while improving customer satisfaction through greater control over waiting time [4].

[Insert Figure 2.6: Customer scanning QR code for queue check-in]

2.5 Transition to the Proposed Solution

The literature reviewed in this chapter establishes that digital queue management technologies significantly improve customer flow organization compared to traditional methods. Key findings include: digital systems reduce wait times through automation [1], real-time information transparency improves customer satisfaction [4], and analytics capabilities enable continuous service optimization [6].

Web platforms extend these benefits by enabling remote queue access and appointment scheduling, while QR code technology provides convenient contactless interaction between physical and digital environments [3], [5].

Based on these concepts, the proposed QONNEXA platform aims to improve queue management at AGB Bank through a web-based system integrating digital tickets, QR code access, and real-time queue monitoring. The platform applies the theoretical principles examined in this chapter to create enhanced experiences for customers, employees, and administrators.

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CHAPTER 3: ANALYSIS AND DESIGN OF THE QONNEXA PLATFORM

3.1 Presentation of the Host Organization

3.1.1 AGB Bank Overview

Gulf Bank Algérie (AGB) is an Algerian commercial bank established in 2003 and operational since March 2004. It is a subsidiary of Burgan Bank Group and a member of KIPCO (Kuwait Projects Company), one of the most prominent business groups in the Middle East [1].

The bank operates with a capital of 20,000,000,000 DZD and has set as its primary mission to contribute to the economic and financial development of Algeria through quality financial services and proximity to clients [2]. AGB Bank has a network of 63 branches operating across 39 wilayas in Algeria, employing approximately 1,074 staff members [3][4]. The headquarters is located in El Biar, Algiers.

[Insert Figure 3.1: AGB Bank logo and headquarters]

Source: AGB Bank official website

3.1.2 Services Offered

AGB Bank offers conventional and Sharia-compliant banking products catering to individual and corporate clients [2].

Table 3.1 – Services Offered by AGB Bank

Service Category	Specific Offerings
Retail Banking	Savings accounts, current accounts, deposit accounts
Consumer Loans	Personal loans, auto loans, mortgage loans

Business Banking	Corporate financing, professional financing, commercial loans
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International Trade	Trade finance, international payment services
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Digital Services	AGB Online, mobile banking, SMS alerts
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Special Services	Sharia-compliant window, Self-Banking, Drive Banking
------------------	--

Source: Compiled from AGB Bank official sources

AGB Bank also provides dedicated services for VIP clients, including dedicated counters and priority queuing.

3.1.3 Customer Flow at AGB Bank Branches

A typical AGB Bank branch has 4 to 8 service counters for cash operations, administrative services, and VIP clients.

[Insert Figure 3.2: Typical AGB Bank branch layout]

Most branches use paper ticket systems. Customers take a numbered ticket and wait until called via audio announcements or display screens.

[Insert Figure 3.3: Current paper ticket dispenser]

Metric	Value
Average daily customers	200 to 500 per branch
Peak hours	9:00 AM – 12:00 PM
Average waiting time	30 to 45 minutes

Current challenges: No waiting time information, customers cannot leave the waiting area, no real-time queue visibility, customer frustration.

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3.2 Problem Statement

3.2.1 Current Queue Management System at AGB Bank

AGB Bank operates a traditional paper-based queue system. Tickets contain only a sequential number without service type distinction. Digital display screens are frequently malfunctioning, show no estimated waiting time, and provide no real-time updates. Staff simply press a "Next" button with no microphone or SMS notifications.

[Insert Figure 3.4: Current paper ticket]

[Insert Figure 3.5: Malfunctioning display screen]

Metric	Value
Average waiting time (peak)	30 to 45 minutes
Maximum waiting time	Up to 4 hours
Customer abandonment rate	10% to 20%

3.2.2 Customer Dissatisfaction

Customer feedback from social media reveals significant dissatisfaction. Customers report waiting four hours for simple transactions and compare the bank's service to postal administration. Long-term customers note service degradation over time, including increased waiting times, reduced staff availability, and frequent technical problems. Multiple customers explicitly state their intention to close accounts or switch to competing banks.

[Insert Figure 3.6: Customer complaints collage]

3.2.3 Lack of Tracking and Consequences

User Type	Limitations
Customers	Cannot see queue position, no wait time estimate, no notifications
Staff	No real-time queue visibility, cannot prepare for customer needs
Management	No statistics, no reports, no performance metrics

Consequences: Customer attrition, reputation damage, lost revenue (10-20% abandonment), competitive disadvantage, inability to improve.

Problem	Required Solution
No queue visibility	Real-time tracking

No wait time information

Estimated wait times

No notifications

SMS/mobile alerts

No data collection

End-of-day statistics

3.3 Proposed Solution: QONNEXA Platform

3.3.1 Overview of the Platform

The **QONNEXA** platform is a web-based queue management solution built with React.js, Node.js, and MySQL. It offers physical and virtual ticketing, real-time queue tracking, and comprehensive analytics. The platform has three access points: Public (QR code), Staff Interface, and VIP Platform.

[Insert Figure 3.7: Platform architecture overview]

Physical Tickets: Pre-printed paper tickets at branch entrances containing ticket number, date, time, bank logo, and branch name. FIFO system.

Virtual Tickets: Customers scan QR code on tablet, select "Create Virtual Ticket," choose service, and receive ticket with position, people ahead, wait time, and counter number.

[Insert Figure 3.8: Tablet interface]

[Insert Figure 3.9: Virtual ticket display]

Queue Tracking: Customers see real-time position, people ahead, and estimated wait time calculated as:

$$ETA = (Position - 1) \times Average\ service\ time / Available\ counters$$

Notifications: In-platform alerts when customer has 3 people ahead, 1 person ahead, or turn arrives.

[Insert Figure 3.10: Queue tracking page]

3.3.2 Platform Features

Feature	Description
Navigation Bar	Queue tracking, FAQ, chatbot, new ticket, satisfaction survey
FAQ	Documents, hours, services, fees, contact information
Chatbot	Answers basic questions (in development, will improve over time)
Satisfaction Survey	4 rating questions (stars) + 1 open comment
VIP Platform	Login, book appointments, priority service (no waiting)
QR Code Integration	Contactless check-in, immediate platform access

[Insert Figure 3.11: Navigation bar]

[Insert Figure 3.12: Satisfaction survey]

[Insert Figure 3.13: VIP booking interface]

3.3.3 Staff, Admin and VIP Interfaces

Interface	Functions
Staff	View waiting tickets, call next customer, start/end service, cancel no-shows, view workload
Admin	Real-time metrics, VIP appointments, counter status, customer feedback
Super Admin	Global dashboard across all branches: active agencies, satisfaction score, traffic by agency, top services; can create agencies, counters, services

[Insert Figure 3.14: Staff dashboard]

[Insert Figure 3.15: Administrator dashboard]

3.4 Functional Requirements

The functional requirements define the specific actions and capabilities that the QONNEXA platform must provide for each user category. These requirements are organized by user type: Client (regular customer), Employee, Administrator, Super Administrator, and VIP Client.

3.4.1 Client Functions

The client (regular customer) can access the QONNEXA platform through two distinct methods: virtual ticket creation via QR code or physical ticket tracking.

Function	Description
Create virtual ticket	Scan QR code → click "Create Virtual Ticket" → select service → receive ticket
Take physical ticket	Select service on tablet → receive printed ticket
Track queue	Enter ticket number → view position, people ahead, wait time
Receive notifications	In-platform alerts (3 ahead, 1 ahead, turn arrives)
Access FAQ	Read answers to common questions
Use chatbot	Ask basic predefined questions
Complete survey	4 rating questions + 1 open comment
Ticket cancellation	Automatic after 2 minutes if customer does not show up

3.4.3 Employee and Administrator Functions

Administrators manage individual branches and have access to a comprehensive dashboard with the following capabilities:

User	Functions
Employee	View waiting list, call next customer, start/end service, cancel no-show tickets, view workload
Admin	View real-time metrics, manage VIP appointments, monitor counter status, adjust counter distribution, view customer feedback

3.4.4 Super Administrator and VIP Functions

User	Functions
Employee	View waiting list, call next customer, start/end service, cancel no-show tickets, view workload
Admin	View real-time metrics, manage VIP appointments, monitor counter status, adjust counter distribution, view customer feedback

3.5 System Architecture

The system architecture of the QONNEXA platform defines the technical structure, components, and communication flows that enable the platform to function efficiently. This section presents the technological choices, system interfaces, backend processing, database design, and communication protocols that form the foundation of the platform.

3.5.1 Technology Stack and System Interfaces

The QONNEXA platform is developed using modern web technologies selected for their performance, scalability, and reliability.

Layer	Technology	Purpose
Frontend	React.js, HTML, CSS	User interface development, component-based architecture, responsive design
Backend	Node.js with Express.js	Server-side logic, API endpoints, request handling
Database	MySQL	Structured data storage, relational data management

The platform is designed as a web application compatible with both mobile and desktop devices, ensuring accessibility across all customer touchpoints.

[Insert Figure 3.21: Technology stack diagram]

System Interfaces

The QONNEXA platform consists of four distinct interfaces, each tailored to specific user needs.

- **Public Platform:** Virtual ticket creation, queue tracking, FAQ, chatbot, satisfaction survey
- **Employee Interface:** View tickets, call next client, start/end service, cancel tickets
- **Admin Dashboard:** Real-time metrics, service management, customer feedback
- **VIP Platform:** Login, appointment booking, appointment management

[Insert Figure 3.22: System interfaces overview]

3.5.2 Backend Processing, Queue Logic and Database

Backend Processing

The backend, built with Node.js and Express.js, handles all server-side logic and API requests.

API Request Flow: When a client creates a ticket, the frontend sends a POST request to the backend:

```
POST /api/tickets{
  "service": "deposit",
  "priority": "normal"
}
```

Backend Processing Steps:

Step	Description
Authentication	JWT for employees/admins; clients remain anonymous
Validation	Verify that the requested service exists
Queue Position Calculation	Query database to determine customer's position
Ticket Creation	Generate ticket (e.g., A045) with position and ETA
JSON Response	Return ticket data in less than 300 milliseconds

Queue Logic and Waiting Time Calculation

The platform implements FIFO (First-In, First-Out) queue discipline. VIP tickets receive priority without disrupting standard queue flow.

Waiting Time Formula:

$ETA = (position - 1) \times avg_service_time / available_counters$

Example: Deposit service (5 min average), position 3, 2 available counters:

- $ETA = (3 - 1) \times 5 / 2 = 5 \text{ minutes}$

Real-Time Queue Updates: When employee clicks "Call Next," the system updates all queue positions instantly and sends real-time updates using WebSocket ([Socket.io](https://socket.io/)).

[Insert Figure 3.23: Queue logic workflow]

Database Design

The platform uses MySQL for structured data storage.

Main Tables:

Table	Description
agencies	Branch information
users	Employees and administrators
services	Service types
tickets	Ticket records
counters	Service counters per agency
appointments	VIP appointments
notifications	Notification records

surveys	Satisfaction survey responses
faqs	Frequently asked questions

Database Features: Primary keys (AUTO_INCREMENT), foreign keys for relationships, indexes for fast queries, triggers for automatic position and ETA calculation.

[Insert Figure 3.24: Database schema diagram]

Communication Between Components

Protocol	Usage
REST API	Ticket creation, authentication, form submissions, data retrieval
WebSocket (Socket.io)	Real-time queue updates, automatic notifications, live dashboards

3.5.3 Communication, Architecture and Components Summary

The platform uses two communication protocols:

Protocol	Usage
REST API	Ticket creation, authentication, form submissions, data retrieval
WebSocket (Socket.io)	Real-time queue updates, automatic notifications, live dashboards

Communication Flow Summary:

Operation	Protocol	Description
Ticket creation	REST API	Client sends request, receives ticket data
Queue position update	WebSocket	Server pushes position changes to all connected clients
Call next customer	REST API + WebSocket	Employee triggers call, system updates all interfaces
Real-time dashboard	WebSocket	Admin dashboard receives live metrics

[Insert Figure 3.26: Communication flow diagram]

System Architecture Overview

The system architecture follows a client-server model with real-time communication capabilities.

Architecture Flow:

1. Client accesses tablet → Frontend sends request via REST API
2. Backend processes request → Queries database → Calculates position and ETA
3. Database returns data → Backend generates ticket
4. Backend sends response to frontend → Client views ticket information
5. Employee clicks "Call Next" → Backend updates queue positions
6. WebSocket pushes real-time updates to all connected clients
7. Clients receive instant notifications without refreshing

Summary of Architecture Components

Component	Technology	Role
Frontend	React.js, HTML, CSS	User interface, responsive design
Backend	Node.js, Express.js	API endpoints, business logic, queue management
Database	MySQL	Data storage, relationships, indexing
Communication	REST API + WebSocket	Request-response and real-time updates
Authentication	JWT	Secure access for employees, admins, and VIPs

CHAPITRE 4 : IMPLEMENTATION OF THE QONNEXA PLATFORM

4.1 Development Environment

4.1.1 Hardware Environment

The development of QONNEXA was carried out using standard computer equipment. Table 4.1 presents the specifications of the machines used during the implementation phase.

Table 4.1 – Development Hardware Specifications

Component	Development Machine
Processor	Intel Core i5 / AMD Ryzen 5
RAM	16 GB
Storage	512 GB SSD
OS	Windows 11 Pro / Ubuntu Linux

4.1.2 Software Environment

The software tools selected for QONNEXA development were chosen based on their reliability, community support, and suitability for building modern web applications [1], [2].

- Visual Studio Code: A lightweight yet powerful source code editor used for both frontend and backend development. It offers extensive extension support for JavaScript, React, and Node.js debugging [3].

- Node.js Runtime: The JavaScript runtime environment that executes server-side code. Node.js uses an event-driven, non-blocking I/O model that makes it efficient and lightweight, ideal for handling concurrent requests in queue management systems [1], [2].
- MySQL Workbench: A unified visual tool for database architects, developers, and administrators. It provides data modeling, SQL development, and comprehensive administration tools for the MySQL Server [4].
- Git: A distributed version control system used for code management, collaboration, and tracking changes throughout the development lifecycle.
- npm (Node Package Manager): The default package manager for Node.js, used to install, manage, and update project dependencies [2].

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4.2 Technologies Used

4.2.1 Frontend Technologies

The frontend of QONNEXA was developed using modern web technologies to create a responsive and intuitive user interface for bank customers, employees, and administrators.

React: A JavaScript library for building user interfaces, chosen for its component-based architecture, virtual DOM for optimal performance, and large ecosystem of libraries and tools [5]. React enables the creation of reusable UI components, making the codebase maintainable and scalable [6].

React Router: A routing library for React that enables navigation between different pages of the application without full page reloads, providing a smooth single-page application (SPA) experience.

Axios: A promise-based HTTP client used to send asynchronous requests from the frontend to the backend API. It handles request and response interceptors, making it easy to manage authentication tokens [7].

CSS3 and Styled Components: Used for styling the application with responsive design, ensuring compatibility across desktop, tablet, and mobile devices.

Chart.js: A JavaScript library for creating interactive charts and graphs, used in the administrator dashboard to visualize queue statistics and performance metrics.

FontAwesome: An icon library providing visual elements for buttons, menus, and navigation components.

4.2.2 Backend Technologies

The backend of QONNEXA handles business logic, data processing, user authentication, and communication with the database.

Node.js: A JavaScript runtime built on Chrome's V8 engine, designed for building scalable network applications. Node.js uses an event-driven, non-blocking I/O model that makes it lightweight and efficient for handling multiple concurrent connections [1], [2]. This architecture is particularly well-suited for queue management systems where multiple clients need real-time updates [8].

Express.js: A minimal and flexible Node.js web application framework that provides a robust set of features for building RESTful APIs. Express handles routing, middleware integration, and request/response processing.

JWT (JSON Web Tokens): Used for secure user authentication and session management. Upon login, a signed token is generated and returned to the client, which includes the user's role and permissions. This token is then sent with each subsequent request to verify identity [9].

bcrypt: A library for hashing passwords before storage in the database, ensuring that user credentials are protected even in case of a data breach.

MySQL: A relational database management system used to store all application data, including users, tickets, services, counters, appointments, and survey responses. MySQL ensures data integrity through relationships and constraints [4], [10].

Sequelize: A promise-based Node.js ORM (Object-Relational Mapping) for MySQL, providing an abstraction layer that simplifies database operations and model management [10].

4.2.3 QR Code Technology

QR (Quick Response) codes are two-dimensional matrix barcodes that can store information such as URLs, text, or other data. The technology was originally developed by Denso Wave in 1994 and has since become widely adopted due to its fast readability and high storage capacity compared to traditional barcodes [11].

QR codes consist of black modules arranged in a square grid on a white background. They can be read by smartphone cameras using QR code reader applications, which decode the information and direct the user to the encoded content [11], [12].

In QONNEXA, QR codes serve multiple functions:

- Contactless check-in: Placed at bank entrances, allowing customers to join the queue by scanning with their smartphones [11].
- Virtual ticket access: Customers who choose virtual tickets can scan the QR code to access their queue status and tracking interface.
- Post-service surveys: QR codes on receipts link to satisfaction surveys for immediate feedback collection.

4.2.4 System Architecture

QONNEXA follows a three-tier client-server architecture:

1. Presentation Tier (Frontend): React application running in the user's browser, handling user interactions and rendering the user interface.
2. Application Tier (Backend): Node.js/Express API handling business logic, authentication, and data processing.
3. Data Tier (Database): MySQL database storing all persistent data.

The frontend and backend communicate through RESTful API endpoints. When a user performs an action (e.g., creating a ticket),

React sends an HTTP request to the Node.js backend, which processes the request, interacts with the database, and returns a JSON response [1].

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4.3 Implementation of the Client Interface

The client interface is the primary point of interaction for bank customers using QONNEXA. It is accessible via QR code scanning and provides two main options: physical ticket and virtual ticket.

4.3.1 QR Code Access and Ticket Creation

When a customer enters an AGB bank branch, they are greeted by a tablet at the entrance displaying a QR code. Scanning this code with a smartphone camera directs them to the QONNEXA web interface.

The interface presents two clear options:

- Physical Ticket: The customer selects a service (e.g., Cash Deposit, Withdrawal, Account Opening) and receives a printed ticket with a unique number and service code.
- Virtual Ticket: The customer selects a service and provides their mobile number. A digital ticket is created, and the customer can track their queue position remotely.

[Insert Figure 4.1: Tablet interface with QR code at bank entrance]

[Insert Figure 4.2: Ticket creation interface with service selection]

The ticket generation logic follows FIFO (First In, First Out) discipline.

The system assigns a sequential number to each ticket, prefixed by a service code (e.g., "CSH-042" for Cash Operations). The ticket is stored in the database with status "waiting", and a confirmation is sent via SMS to the customer's mobile number.

[Insert Figure 4.3: Ticket confirmation screen with QR code for tracking]

4.3.2 Queue Tracking Interface

For customers with virtual tickets, the QONNEXA platform provides a real-time tracking interface accessible through the same QR code or via a link sent by SMS.

The tracking interface displays:

- Current position: "You are number 7 in the queue"
- Estimated waiting time: "Approximately 12 minutes until your turn"
- Queue status: Number of customers ahead and active counters
- Notifications toggle: Option to receive SMS/email updates

The estimated waiting time is calculated dynamically using the formula:

text

Estimated Wait Time = (Position in Queue) × (Average Service Time per Service Type)

Average service times are maintained per service type based on historical data, ensuring increasingly accurate predictions over time.

[Insert Figure 4.4: Queue tracking interface on mobile phone]

4.3.3 Notification System

The notification system keeps customers informed throughout their journey. Three types of notifications are implemented:

1. Ticket Confirmation: Sent immediately after ticket creation, confirming service type, ticket number, and initial queue position.
2. Near-turn Alert: Sent when the customer has 3 tickets ahead, giving them time to return to the bank.
3. Turn Ready Alert: Sent when the customer's ticket is called, indicating which counter to approach.

Notifications are delivered via SMS using a third-party SMS gateway integrated with the Node.js backend.

[Insert Figure 4.5: SMS notification examples]

4.3.4 Satisfaction Survey

After service completion, customers are invited to rate their experience. A QR code on the printed receipt (or a link in the final notification) directs to a short survey.

The survey collects:

- Overall satisfaction: 1-5 star rating
- Waiting time satisfaction: 1-5 star rating

- Staff helpfulness: 1-5 star rating
- Optional comments: Text field for feedback

Survey responses are stored in the database and aggregated in the administrator dashboard, providing valuable insights for service improvement.

[Insert Figure 4.6: Satisfaction survey interface]

4.4 Implementation of the Employee Interface

Bank employees access the QONNEXA employee interface through a secure login page. Each employee has credentials and a role determining their access level.

4.4.1 Employee Dashboard

Upon login, the employee dashboard displays:

- Current queue: List of waiting customers with ticket numbers and service types
- Next customer: The next ticket to be called
- Active counter: The employee's assigned counter number
- Today's statistics: Number of customers served, average service time

[Insert Figure 4.7: Employee dashboard interface]

4.4.2 Queue Management Functions

Employees can perform the following actions:

Call Next Customer: When ready to serve, the employee clicks "Call Next". The system selects the next customer based on queue discipline

(FIFO with priority handling for VIP appointments). A notification is sent to the customer, and the ticket number appears on digital displays.

Complete Service: After serving the customer, the employee marks the service as complete. This updates queue statistics, records service duration, and triggers the satisfaction survey for the customer.

Transfer Customer: If a customer needs a different service, the employee can transfer them to another queue without losing their position.

No-show: If a called customer does not appear after three announcements, the employee can mark them as no-show and move to the next customer.

[Insert Figure 4.8: Employee queue control panel]

The backend API handles these operations using Express routes. For example, calling the next customer involves:

1. Querying the database for waiting tickets in FIFO order
 2. Updating ticket status to "called"
 3. Sending notification via SMS
 4. Updating digital displays
-

4.5 Administrator Dashboard

The administrator dashboard provides a comprehensive view of queue operations across the entire branch. It is accessible only to users with administrator privileges.

4.5.1 Overview Dashboard

The main dashboard displays:

- Real-time metrics: Current queue length, average wait time, number of active counters
- Today's statistics: Customers served, peak hours, average service time
- Alerts: Unusual wait times, system issues, low ticket paper

[Insert Figure 4.9: Administrator dashboard overview]

4.5.2 Statistics and Reporting

The statistics module collects and visualizes data to support decision-making. Key reports include:

- Hourly/Daily/Weekly trends: Customer volume patterns
- Service type distribution: Which services are most requested
- Employee performance: Average service time by employee
- Customer satisfaction: Survey results over time
- Abandonment rate: Customers who left before service

Data is presented through interactive charts created with Chart.js, allowing administrators to filter by date range, service type, or employee.

[Insert Figure 4.10: Statistics charts and graphs]

4.5.3 Service Management

Administrators can configure system parameters:

- Service types: Add, edit, or disable services offered
- Service priorities: Assign priority levels (e.g., VIP, urgent)
- Estimated times: Set baseline service duration estimates
- Counter assignment: Map services to specific counters

[Insert Figure 4.11: Service configuration interface]

4.5.4 User Management

The administrator manages all system users:

- Create accounts: Add new employees with appropriate roles
- Modify roles: Change access permissions
- Disable accounts: Remove access for former employees
- Reset passwords: Assist users with login issues

[Insert Figure 4.12: User management interface]

4.6 VIP Appointment Platform

VIP clients have access to a dedicated web portal for scheduling appointments remotely. This feature enhances the premium banking experience by providing priority service.

4.6.1 Appointment Booking Interface

The VIP portal allows clients to:

- View available slots: Calendar showing open time slots for each service
- Select service: Choose the type of transaction needed
- Book appointment: Reserve a specific time
- Receive confirmation: Email and SMS with appointment details
- Modify/Cancel: Change or cancel existing appointments

[Insert Figure 4.13: VIP appointment booking calendar]

4.6.2 Priority Queue Integration

When a VIP client arrives at the branch, their appointment is integrated with the main queue system:

- Identity verification: Client provides name or confirmation number
- Check-in: Marks the client as arrived for their appointment

- Priority handling: VIP appointments receive priority over walk-in customers while maintaining fairness

The system uses a priority queue algorithm that ensures VIP customers are served within their scheduled time window.

4.6.3 Reminder System

Automatic reminders are sent to VIP clients:

- 24 hours before: Email reminder with appointment details
 - 1 hour before: SMS reminder
 - Upon arrival: Instructions for check-in upon reaching the branch
-

4.7 System Testing

4.7.1 Unit Testing

Each component of QONNEXA was tested individually to ensure correct functionality. Tests included:

- Ticket generation: Verifying unique ticket numbers, correct formatting, and service code assignment
- Queue ordering: Confirming FIFO logic works correctly and priority handling for VIP appointments
- Notification delivery: Testing SMS sending with mock providers
- Authentication: Validating login, JWT token generation, and role-based access control

4.7.2 Integration Testing

Integration tests verified that all system components work together:

- Frontend-Backend communication: API endpoints return correct responses with appropriate status codes
- Database interactions: Data is properly stored and retrieved with correct relationships

- External services: SMS gateway integration functions reliably with error handling

4.7.3 User Acceptance Testing

A group of users tested the system in a simulated environment:

- Customers: 10 participants tested ticket creation and tracking
- Employees: 5 bank staff tested queue management functions
- Administrator: 1 admin tested dashboard and configuration

Feedback was collected and used to make final adjustments, including:

- Improved visual clarity of the queue tracking interface
- Added confirmation messages for all user actions
- Enhanced error handling for invalid inputs

[Insert Figure 4.14: Test results summary]

4.8 Discussion of Results

The implementation of QONNEXA demonstrates how digital queue management can transform the customer experience at AGB Bank.

4.8.1 Benefits Achieved

- Reduced physical congestion: Virtual tickets allow customers to wait elsewhere, decreasing crowding in the banking hall
- Improved customer satisfaction: Real-time information and notifications reduce anxiety about waiting
- Enhanced efficiency: Automated queue management frees employees from manual organization tasks
- Data-driven decisions: Analytics provide insights for optimizing service delivery
- VIP experience: Priority handling and appointment scheduling enhance premium customer satisfaction

4.8.2 Limitations

- Internet dependency: The system requires stable internet connectivity for both customers and staff
- Smartphone requirement: Virtual ticket features require customers to have mobile phones
- Learning curve: Some employees needed training to use the new system effectively
- Initial setup: QR code placement and tablet installation required physical configuration

4.8.3 Future Improvements

Potential enhancements for future versions include:

- Mobile application: Native app for iOS and Android with push notifications
- AI prediction: Machine learning to predict wait times more accurately based on historical patterns
- Biometric check-in: Fingerprint or facial recognition for VIP clients
- Multi-branch integration: Unified queue management across all AGB branches
- Advanced analytics: Predictive staffing recommendations based on forecasted customer volume

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C'est prêt ! Le Chapitre 4 est complet. Tu veux qu'on passe au Chapitre 3 (que fait ton binôme) ou on ajoute quelque chose ?